WORKED SOLUTION

STEP III 1999, Question 13

Probability and Statistics • Bayes with Continuous Prior

STEP III 1999, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q13

The key idea

There are two layers of randomness: first the size X of my portion, then where the currants land. Once we *condition* on $X = x$, each currant is in my portion with probability x, independently of the others, so the number of currants is binomial. After that, part (i) is the law of total expectation and part (ii) is Bayes' theorem with a continuous prior.

Solution

First we find A. A density must integrate to 1:

$$\int_0^1 Ax\,\mathrm{d}x = \frac{A}{2} = 1 \quad \Longrightarrow \quad A = 2,$$

so X has density $\mathrm{f}(x) = 2x$ on $[0, 1]$.

Let N be the number of currants in my portion. We model "randomly placed" in the natural way: each currant's position is uniformly distributed through the cake, independently of the other currants and of how much cake I take. If I take a proportion x of the cake, each currant therefore lies in my portion with probability x, independently of the others, so

$$N \mid (X = x) \sim \mathrm{B}(4, x).$$

Part (i)

Given $X = x$, the binomial mean is $\mathrm{E}(N \mid X = x) = 4x$. Averaging over X (the law of total expectation),

$$\begin{aligned}\mathrm{E}(N) &= \int_0^1 \mathrm{E}(N \mid X = x)\,\mathrm{f}(x)\,\mathrm{d}x = \int_0^1 4x \cdot 2x\,\mathrm{d}x \\ &= 8\int_0^1 x^2\,\mathrm{d}x = \frac{8}{3}.\end{aligned}$$

Equivalently, $\mathrm{E}(N) = 4\,\mathrm{E}(X)$ with $\mathrm{E}(X) = \frac{2}{3}$. The expected number of currants in my portion is $\frac{8}{3}$.

Part (ii)

We want $\mathrm{P}(X > \frac{1}{2} \mid N = 4)$. Given $X = x$, all four currants are in my portion with probability x^4.

> **Ask yourself** How do we use Bayes' theorem when X is continuous and $\mathrm{P}(X = x) = 0$?

We replace the sums in Bayes' theorem by integrals against the density. A slice $[x, x + \mathrm{d}x]$ of possible portion sizes has probability $2x\,\mathrm{d}x$, and given that slice the chance of all four currants is x^4. Adding up over slices,

$$\mathrm{P}(N = 4) = \int_0^1 x^4 \cdot 2x\,\mathrm{d}x = 2\int_0^1 x^5\,\mathrm{d}x = \frac{2}{6} = \frac{1}{3}.$$

In the same way, keeping only the slices with $x > \frac{1}{2}$,

$$\mathrm{P}(X > \tfrac{1}{2} \text{ and } N = 4) = \int_{1/2}^1 2x^5\,\mathrm{d}x = \frac{1}{3}\Big(1 - \frac{1}{64}\Big) = \frac{1}{3}\cdot\frac{63}{64}.$$

Therefore

$$\mathrm{P}(X > \tfrac{1}{2} \mid N = 4) = \frac{\frac{1}{3}\cdot\frac{63}{64}}{\frac{1}{3}} = \frac{63}{64}.$$

Another way to see it: given $N = 4$, the conditional density of X is proportional to $x^4 \cdot 2x$, so it is $6x^5$ on $[0,1]$. Its distribution function is x^6, and $1 - (\frac{1}{2})^6 = \frac{63}{64}$.

Checks

- The answer to (ii) is very close to 1, which makes sense: finding every currant is strong evidence that I took a big piece. The prior probability $\mathrm{P}(X > \frac{1}{2}) = 1 - \frac{1}{4} = \frac{3}{4}$ has been pushed up to $\frac{63}{64} \approx 0.984$.
- $\frac{8}{3}$ lies between 0 and 4, and it is more than 2, as it should be: the density $2x$ favours large portions.
- A Monte Carlo simulation with 10^6 trials (taking $X = \sqrt{U}$ with U uniform, which has density $2x$, then placing four currants) gave $\mathrm{E}(N) \approx 2.667$, $\mathrm{P}(N = 4) \approx 0.333$ and a conditional probability ≈ 0.9846, against $\frac{63}{64} = 0.984375$.

Common slips

- Forgetting to find A, or taking $A = 1$. Then f isn't a density, and the expectation in (i) comes out wrong (although A cancels in (ii)).

- Computing $\mathrm{P}(N = 4 \mid X > \frac{1}{2})$ instead of $\mathrm{P}(X > \frac{1}{2} \mid N = 4)$. The conditioning is the other way round.
- Using $(\frac{1}{2})^4$ in place of the integral: the portion size is random, so we must average x^4 against the density rather than plug in a single value.

Beyond STEP

This is Bayesian inference in miniature: the density $2x$ is a *prior* for the unknown proportion, the binomial is the *likelihood*, and $6x^5$ is the *posterior*. Both prior and posterior are beta distributions, an example of a conjugate prior for binomial data, and the same calculation underlies Laplace's rule of succession. Blitzstein and Hwang, *Introduction to Probability*, treats the beta–binomial story and conditioning on continuous random variables very clearly.

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